Code:

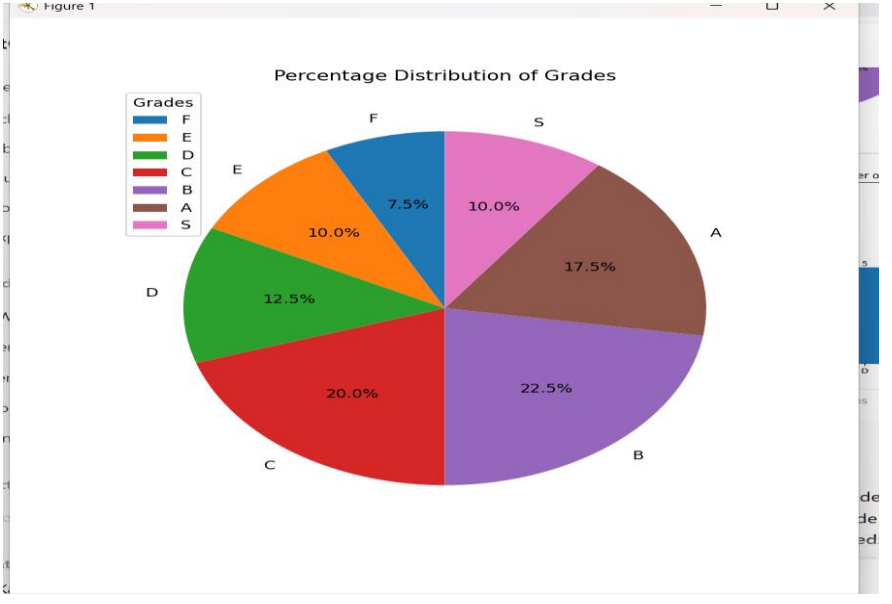
```
import matplotlib.pyplot as plt
grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']
students = [3, 4, 5, 8, 9, 7, 4]
total = sum(students)
percentages = [(x / total) * 100 for x in students]
plt.figure(figsize=(7, 7))
plt.pie(
    students,
    labels=grades,
    autopct='%1.1f%%',
    startangle=90
)
plt.title('Percentage Distribution of Grades')
```

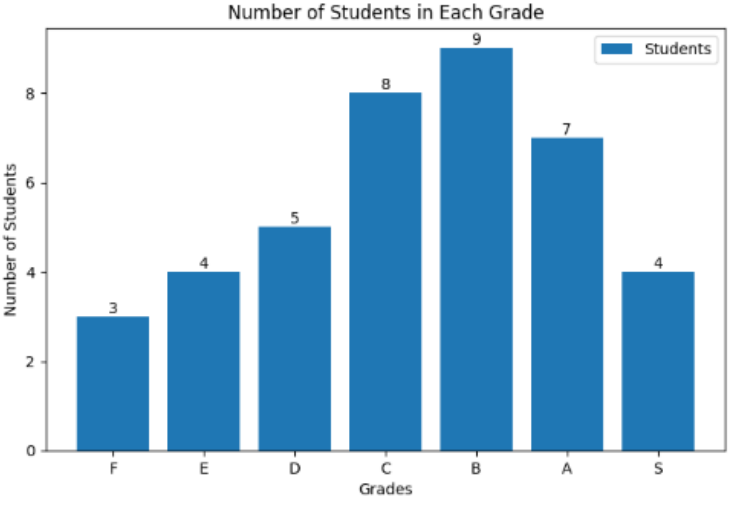
```
plt.legend(grades, title='Grades')
plt.show()
plt.figure(figsize=(8, 5))
bars = plt.bar(grades, students, label='Students')
plt.xlabel('Grades')
plt.ylabel('Number of Students')
plt.title('Number of Students in Each Grade')
plt.legend()
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        str(int(height)),
```

```
        ha='center',
        va='bottom'
    )
plt.show()
max_index = students.index(max(students))
min_index = students.index(min(students))
print("Maximum students:", grades[max_index],
      "-", students[max_index])
print("Minimum students:", grades[min_index],
      "-", students[min_index])
combined = students[5] + students[6]
combined_percentage = (combined / total) * 100
print("Students with grades A and S:", combined)
```

print("Combined percentage:", round(combined_percentage, 2), "%")

OUTPUT:




Number of Students in Each Grade
Students
Number of Students
Grades
0
2
4
6
8
F
E
D
C
B
A
S
3
4
5
8
9
7
4